

Guide: What Sampled Contraction Would Prove

Collatz proof project

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The exact result

Lean now proves that the following two claims are equivalent: every positive Collatz orbit is bounded; and every positive seed two modulo nine eventually has a coefficient below one at a later visit to that same progression. Both claims remain unproved.

Why restricting the observations works

Suppose the coefficient never contracts at any of those visits. Between consecutive visits, at most six even steps occur. Even steps halve the coefficient and odd steps increase it, so at every intervening time it remains at least one sixty-fourth. Our previous exact correction theory then shows that such a positive seed must have an unbounded orbit.

Conversely, on any hypothetical unbounded orbit the inverse coefficient tends to zero. Look only at visits to two modulo nine and choose a visit where this inverse coefficient is largest. Relative to that starting point, every later sampled coefficient is at least one. This supplies both the required residue and sampled noncontraction in the same tail.

What remains unresolved

The selected tail need not have coefficients at least one at every time; the proved intervening bound is one sixty-fourth. The distinction matters when applying earlier lemmas that require full noncontraction. We still need a proof that every seed in the progression eventually contracts at a visit. The equivalence identifies that as the full nondivergence problem. It does not eliminate nontrivial cycles. The finite experiment about descent at the first sampled contraction remains a separate, unproved candidate. No proof of Collatz is claimed.

Verification

From `lean/`, run `lake build Collatz.SampledContraction`. The repository's selected theorem audit checks the four new declarations and their axiom dependencies. The proof uses existing exact theorems, without importing an experimental result as an assumption.